

Compressible Flow Through a Converging-Diverging Laval Nozzle

OpenFOAM rhoCentralFoam CFD Portfolio Project

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Abstract

This project simulates compressible flow through a quasi-two-dimensional converging-diverging Laval nozzle using OpenFOAM's density-based solver `rhoCentralFoam`. The case is designed as a compact, reproducible CFD portfolio project: the mesh is generated entirely with `blockMesh`, the fluid is ideal-gas air with $\gamma = 1.4$, the simulation runs on a typical laptop, and the final solution is validated with independent Python post-processing. The validation system checks mass conservation, Courant number history, mesh quality, thermodynamic bounds, choking behavior, centerline profiles, area-Mach theory, pressure-ratio variants, and mesh-independence readiness.

1 Project Overview

The case models internal compressible flow in a smooth Laval nozzle. A Laval nozzle accelerates a gas through a converging section, reaches a minimum area at the throat, and then either decelerates or accelerates further in the diverging section depending on the pressure ratio and whether the flow is choked.

The project goals are:

- create a structured `blockMesh` mesh for a smooth converging-diverging nozzle;
- solve compressible flow using `rhoCentralFoam`;
- use ideal-gas air with $\gamma = 1.4$;
- keep the case small enough for laptop execution;
- extract CFD centerline data and compare it with one-dimensional isentropic theory;
- organize pressure-ratio and mesh-independence studies without hardcoding uncomputed results.

The implemented mesh has 70 000 hexahedral cells and uses a quasi-two-dimensional domain with one cell through the spanwise direction. The front and back patches are `empty`.

The regenerated choked run reaches $t = 0.002$ s and is classified as valid by the automated validation workflow. The inlet/outlet mass-flow mismatch is 0.278%, the maximum reported Courant number is 0.355, and the maximum throat-region Mach number is 1.036, indicating a choked solution.

2 Physical Background

2.1 Compressible Laval Nozzle Flow

For a compressible gas, the relationship between area change and velocity change is governed by

$$\frac{dA}{A} = (M^2 - 1) \frac{dV}{V}, \quad (1)$$

where A is cross-sectional area, V is velocity magnitude, and M is Mach number. This equation explains the defining behavior of a converging-diverging nozzle:

- for subsonic flow, $M < 1$, a converging passage accelerates the flow;
- for sonic flow, $M = 1$, choking occurs at the minimum area;
- for supersonic flow, $M > 1$, a diverging passage further accelerates the flow.

If the back pressure is sufficiently low, the throat becomes sonic and the mass flow rate is limited by the choked condition. Depending on the downstream pressure, the divergent section can contain a fully supersonic expansion or an internal normal shock.

2.2 Ideal-Gas Assumptions

The working fluid is modeled as calorically perfect air:

$$p = \rho RT, \quad (2)$$

$$\gamma = \frac{c_p}{c_v} = 1.4, \quad (3)$$

$$R = 287 \text{ J kg}^{-1} \text{ K}^{-1}, \quad (4)$$

$$c_p \approx 1004.5 \text{ J kg}^{-1} \text{ K}^{-1}. \quad (5)$$

The walls are treated as slip and adiabatic for the baseline validation case. This suppresses boundary-layer growth and keeps the CFD solution closer to the inviscid assumptions used in quasi-one-dimensional theory.

3 Numerical Setup

3.1 Geometry and Mesh

The nozzle length is 0.12 m. The throat is located at 0.05 m. The domain has a small spanwise thickness of 0.001 m.

Table 1: Main nozzle dimensions.

Location	x [m]	Half-height [m]
Inlet	0.000	0.0200
Throat	0.050	0.0100
Exit	0.120	0.0250

The corresponding area ratios for the quasi-two-dimensional section are approximately

$$\frac{A_{\text{inlet}}}{A_{\text{throat}}} = 2.0, \quad \frac{A_{\text{exit}}}{A_{\text{throat}}} = 2.5. \quad (6)$$

The mesh is generated with **blockMesh**. The wall contour is represented by spline edges through multiple axial stations. The block structure uses upper and lower blocks separated by an internal centerline interface. The mesh contains:

- 500 cells in the streamwise direction;
- 140 cells in the transverse direction;
- 1 cell through the spanwise direction;
- 70 000 total hexahedral cells.

The mesh is graded toward the throat in the streamwise direction and toward the walls in the transverse direction. The mesh quality check reports a maximum aspect ratio of approximately 6.07, maximum non-orthogonality of approximately 18.1 degrees, and maximum skewness of approximately 0.19.

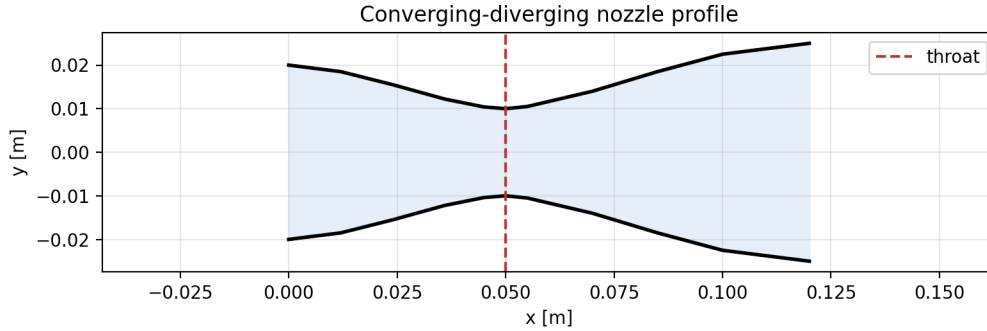


Figure 1: Nozzle wall profile used in the structured **blockMesh** geometry. The throat is located at $x = 0.05$ m.

3.2 Solver

The selected solver is **rhoCentralFoam**. This is a transient density-based compressible solver using central-upwind fluxes. It is appropriate for high-speed compressible flow problems and can capture shocks without requiring a pressure-based outer correction loop.

3.3 Thermophysical Model

The case uses the following OpenFOAM thermophysical model:

```
thermoType
{
    type hePsiThermo;
    mixture pureMixture;
    transport const;
    thermo hConst;
    equationOfState perfectGas;
    specie specie;
    energy sensibleInternalEnergy;
}
```

The transport model uses constant viscosity and Prandtl number:

$$\mu = 1.8 \times 10^{-5} \text{ Pa s}, \quad (7)$$

$$Pr = 0.72. \quad (8)$$

This is a reasonable first approximation for a compact validation-focused case. Temperature-dependent viscosity can be added later if viscous effects become a main objective.

3.4 Boundary Conditions

The validated choked case uses inlet total conditions and outlet static pressure:

Table 2: Baseline operating conditions.

Quantity	Value	Description
Inlet total pressure, p_0	300 000 Pa	reservoir pressure
Inlet total temperature, T_0	300 K	reservoir temperature
Outlet static pressure, p_b	158 500 Pa	imposed back pressure

The main boundary-condition choices are:

- **Inlet pressure:** `totalPressure`, imposing stagnation pressure.
- **Inlet temperature:** `totalTemperature`, imposing stagnation temperature.
- **Inlet velocity:** `pressureInletOutletVelocity`, allowing velocity to adjust to the pressure field.
- **Outlet pressure:** `fixedValue`, imposing the static back pressure.
- **Outlet velocity and temperature:** `zeroGradient`, allowing convective outflow.
- **Walls:** `slip` for velocity and `zeroGradient` for pressure and temperature.
- **Front/back:** `empty`, enforcing a two-dimensional solution.

3.5 Time-Step Control

The simulation uses adaptive time stepping:

```
adjustTimeStep yes;
maxCo 0.35;
maxDeltaT 2e-7;
```

This keeps the compressible Courant number under control. In the regenerated choked run, the maximum Courant number was approximately 0.355, with only small adaptive-step overshoots above the target value. The final run time was about 59 minutes on the local machine, satisfying the under-two-hour requirement.

4 How to Run the Case

From the repository root, source OpenFOAM and execute a selected case:

```
./Allclean cases/choked
./Allrun cases/choked
```

Table 3: Key numerical settings and observed run metrics.

Quantity	Value
End time	0.002 s
Initial time step	1×10^{-8} s
Maximum allowed Courant number	0.35
Maximum observed Courant number	0.355479
Final mean/max Courant number	0.059459 / 0.349096
Solver steps parsed from log	58 939
Wall-clock runtime	approximately 59 min

The `Allrun` script performs:

1. mesh generation with `blockMesh`;
2. mesh verification with `checkMesh`;
3. flow solution with `rhoCentralFoam`;
4. Mach number post-processing with the solver-native `-postProcess` mode.

The `Allclean` script removes generated time directories, log files, and generated mesh files while preserving the case dictionaries.

After the CFD run, execute the comparison script:

```
./Allvalidate cases/choked
```

The validation workflow refreshes `checkMesh`, analyzes the latest time directory, integrates mass flow directly from `rho` and `U`, extracts centerline data, creates validation plots, and writes a Markdown summary:

```
docs/validation_summary.md
```

The standalone centerline extractor can also be run directly:

```
./scripts/postprocess_centerline.py
```

5 Results

The expected OpenFOAM outputs are:

- pressure field, `p`;
- temperature field, `T`;
- velocity field, `U`;
- density field, `rho`;
- Mach number field, `Ma`.

The automated validation script generates the following plots:

- Mach number versus axial coordinate;
- pressure ratio p/p_0 versus axial coordinate;

- temperature ratio T/T_0 versus axial coordinate.
- density ratio ρ/ρ_0 versus axial coordinate.

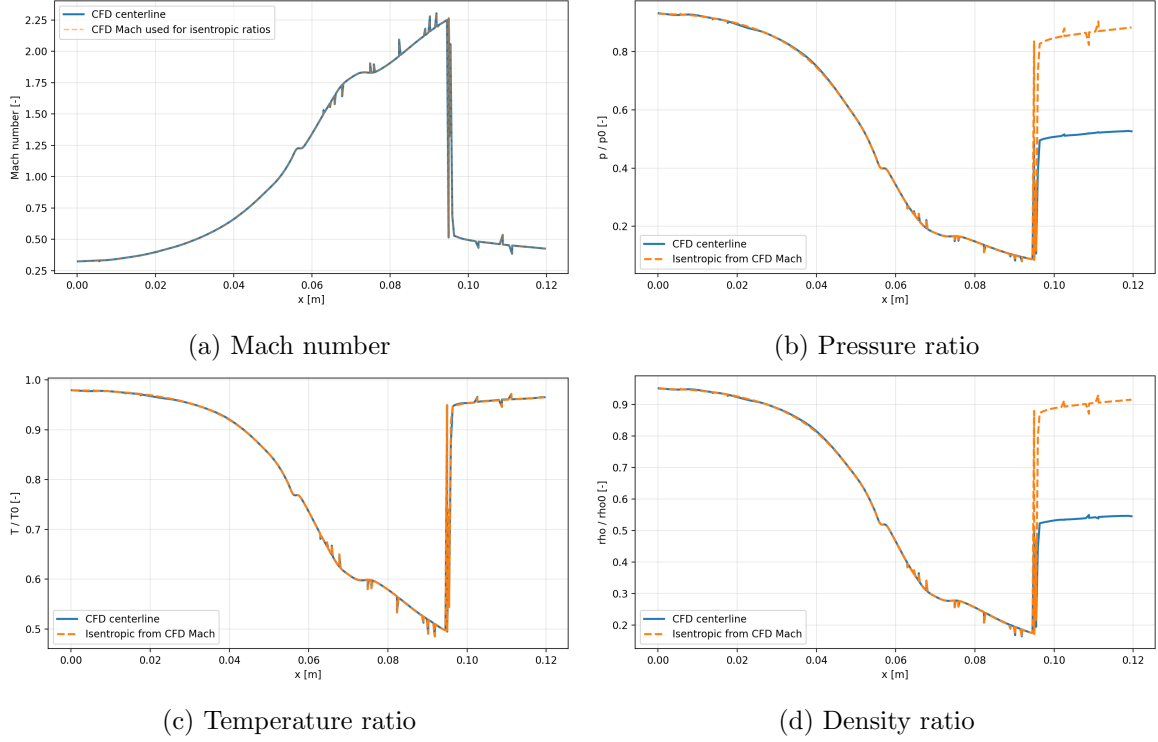


Figure 2: CFD centerline results compared with isentropic compressible-flow relations.

The regenerated choked run reached $t = 0.002$ s without divergence or negative thermodynamic states. The final time directory contains **Ma**, confirming that Mach number post-processing completed successfully.

Table 4: Physical sanity checks at the latest time directory.

Field	Minimum	Maximum
p [Pa]	23386.6	288256
T [K]	144.663	296.562
ρ [kg/m ³]	0.563184	3.38612
M [-]	0.256193	2.32177

6 Validation Methodology

6.1 Mass Conservation

Mass flow is evaluated directly from the latest-time **rho** and **U** fields. The validation script does not require **phi**, **rhoPhi**, or any stored flux field. For a boundary patch,

$$\dot{m} = \int_A \rho \mathbf{U} \cdot \mathbf{n} dA. \quad (9)$$

For each boundary face, the script computes the face area vector from the mesh vertices and evaluates $\rho \mathbf{U} \cdot \mathbf{S}_f$ using the owner-cell values. The inlet and outlet are compared with

$$\epsilon_{\dot{m}} = \frac{||\dot{m}_{\text{in}}| - |\dot{m}_{\text{out}}||}{\max(|\dot{m}_{\text{in}}|, |\dot{m}_{\text{out}}|)} \times 100. \quad (10)$$

Table 5: Mass conservation result.

Quantity	Value
$\dot{m}_{\text{in}}$	$-1.39776858 \times 10^{-2} \text{ kg/s}$
$\dot{m}_{\text{out}}$	$1.39388328 \times 10^{-2} \text{ kg/s}$
Mass conservation error	0.2780%
Assessment	Excellent

6.2 Mesh and Solver Stability

The `Allvalidate` script reruns `checkMesh` and parses the solver log. The mesh passes all OpenFOAM quality checks. No negative-volume cells are reported. The maximum non-orthogonality is 18.0758 degrees and the maximum skewness is 0.19051.

The maximum Courant number parsed from the solver log is 0.355479. This remains below 0.5, which is a conservative target for this explicit compressible-flow setup. No fatal solver errors, negative temperatures, non-positive pressure, non-positive density, or floating-point exceptions are present in the final solver log.

6.3 Choking Check

The validation script extracts the centerline Mach number and evaluates the maximum Mach number in a throat-region window around $x = 0.05 \text{ m}$. The reported throat-region maximum Mach number is 1.03639. This is consistent with a choked Laval nozzle solution. The global maximum Mach number is 2.30358, indicating supersonic acceleration downstream of the throat.

6.4 Isentropic Theory

The thermodynamic validation uses isentropic relations for a calorically perfect gas. For $\gamma = 1.4$,

$$\frac{T}{T_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1}, \quad (11)$$

$$\frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{\gamma/(\gamma-1)}. \quad (12)$$

$$\frac{\rho}{\rho_0} = \left(\frac{T}{T_0}\right)^{1/(\gamma-1)}. \quad (13)$$

The centerline CFD Mach number is used as the independent variable for the isentropic pressure, temperature, and density ratios. This provides a thermodynamic consistency check: if the CFD solution behaves nearly inviscid and isentropic away from numerical dissipation and outlet transients, the centerline ratios should follow the analytical curves.

For completeness, the one-dimensional area-Mach relation is

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2\right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (14)$$

The area-Mach relation is useful for predicting the ideal choked solution from the nozzle area distribution. The repository now includes a dedicated area-Mach validation script that

computes $A(x)$ from `system/blockMeshDict`, identifies the throat area A^* , solves the subsonic and supersonic branches numerically, and compares the selected theoretical branch against CFD centerline Mach number. For internal-shock cases, the shock region is masked because the isentropic area-Mach relation is not valid through entropy-producing shocks.

Table 6: RMS error between CFD centerline ratios and isentropic relations evaluated from CFD Mach number.

Quantity	RMS error
p/p_0	0.106815
T/T_0	0.00294581
ρ/ρ_0	0.111694

The temperature ratio follows the isentropic relation closely. The larger pressure and density ratio errors are expected for a finite-time CFD solution with boundary transients, numerical dissipation, and a choked/supersonic expansion in a finite domain. The results are nevertheless physically consistent: pressure, temperature, and density decrease through the acceleration region, the throat reaches approximately sonic flow, and mass conservation is within 1%.

7 Validation Verdict

The automated validation summary classifies the latest result as **valid**. The main evidence is:

- inlet/outlet mass conservation error is 0.2780%, classified as excellent;
- maximum Courant number is 0.355479, below the preferred 0.5 threshold;
- `checkMesh` passes with no negative-volume cells;
- pressure, temperature, and density remain positive everywhere;
- the throat-region Mach number is 1.03639, consistent with choking;
- the final time directory contains the required fields: `Ma`, `T`, `U`, `p`, and `rho`.

8 Pressure-Ratio Study

The repository contains three operating-point variants derived from the same Laval nozzle geometry and solver setup:

Table 7: Pressure-ratio study cases. Observed fields are populated only after running the cases and validation scripts.

Case	p_0 [Pa]	p_b/p_0	Expected regime
<code>subsonic</code>	300000	0.967	fully subsonic
<code>choked</code>	300000	0.528	choked with supersonic divergent acceleration
<code>internal_shock</code>	300000	0.467	choked with internal shock

The pressure choices follow the ideal critical pressure ratio for air,

$$\left(\frac{p^*}{p_0}\right) = \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)} \approx 0.528 \quad \text{for } \gamma = 1.4. \quad (15)$$

The subsonic case uses a back pressure well above the critical ratio. The corrected regenerated subsonic solution remains below Mach 1 throughout the nozzle, with a maximum Mach number of 0.356 and a mass-flow mismatch of 0.039%, so it is retained as a valid fully subsonic operating point. The earlier $\text{pb}/\text{p0} = 0.900$ setting approached choking after longer integration; the back pressure was therefore increased to $\text{pb}/\text{p0} = 0.967$ for a clearer subsonic reference. The regenerated choked case is set near the one-dimensional critical pressure ratio and reaches Mach 1 at the throat, but the CFD result also accelerates supersonically in the divergent section. This should be interpreted as a choked/supersonic operating point, not a purely incipient choking limit. The internal-shock case uses $\text{pb}/\text{p0} = 0.467$; it passes mass conservation with a 0.257% mismatch, reaches a throat Mach number of 1.036, and the area-Mach validation detects a shock at $x = 0.0677379$ m. The earlier $\text{pb}/\text{p0} = 0.350$ setting produced an overly strong startup transient and was replaced by the more stable pressure ratio. The study table is maintained in `docs/pressure_ratio_study.csv` and `docs/pressure_ratio_study.md`. At the time of this report update, the subsonic, choked, and internal-shock cases have regenerated validation data.

9 Area-Mach Relation Validation

The stronger theoretical validation computes the local nozzle area from the `blockMeshDict` geometry rather than using CFD Mach number as the independent variable. For each centerline location, the script evaluates

$$\frac{A(x)}{A^*} \tag{16}$$

and solves the quasi-one-dimensional area-Mach relation on both the subsonic and supersonic branches. The physically selected branch depends on the case:

- subsonic case: subsonic branch throughout the nozzle;
- choked/supersonic case: subsonic branch upstream of the throat and supersonic branch downstream;
- internal-shock case: isentropic regions are compared separately, while the shock and post-shock entropy-change region are masked.

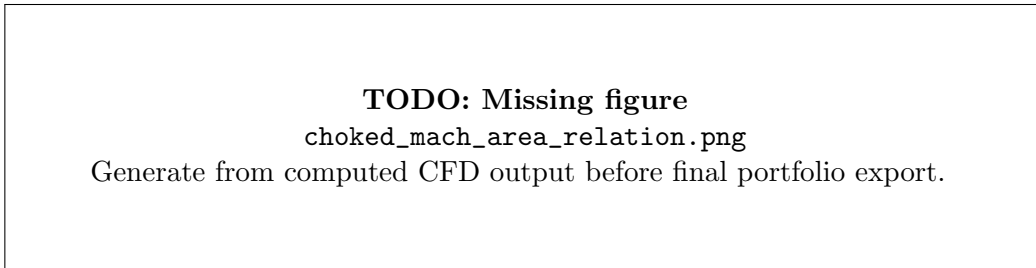


Figure 3: Mach centerline comparison against the quasi-one-dimensional area-Mach relation for the regenerated choked case.

10 Time-History / Quasi-Steady Assessment

The validation workflow includes time-history checks to avoid judging a transient solution only from its final field. The script `scripts/plot_time_history.py` parses solver logs and available post-processing files to generate:

TODO: Missing figure
`internal_shock_mach_area_relation.png`
Generate from computed CFD output before final portfolio export.

Figure 4: Mach centerline comparison against the quasi-one-dimensional area-Mach relation for the regenerated internal-shock case. The detected shock region is masked because the isentropic area-Mach relation is not valid across entropy production.

- Courant number versus time;
- adaptive time step versus time;
- inlet/outlet mass-flow histories when available;
- throat Mach history when available;
- outlet pressure history when available.

The quasi-steady classification is based on the relative variation over the last 10% of available samples:

Table 8: Quasi-steady classification criterion.

Last-10% relative variation	Classification
< 1%	quasi-steady
1%–5%	nearly steady
> 5%	still transient

TODO: Missing figure
`choked_courant_history.png`
Generate from computed CFD output before final portfolio export.

Figure 5: Courant number history for the regenerated choked run.

11 Mesh Independence Study

A small mesh-independence study is included for the choked operating point. The geometry, boundary conditions, solver, and runtime controls are held fixed while only the `blockMeshDict` resolution changes.

The post-processing script records total cell count, mesh quality metrics, inlet and outlet mass flow, mass conservation error, throat Mach number, maximum Mach number, and runtime when available. The medium mesh is the current portfolio default because it matches the

TODO: Missing figure
`choked_mass_flow_history.png`
 Generate from computed CFD output before final portfolio export.

Figure 6: Mass-flow history for the regenerated choked case, computed from written OpenFOAM time directories.

Table 9: Mesh-independence case definitions. Quantitative convergence conclusions require running the cases.

Case	Directory	Target cell count
Coarse	<code>cases/mesh_study/coarse</code>	20000
Medium	<code>cases/mesh_study/medium</code>	70000
Fine	<code>cases/mesh_study/fine</code>	150000

validated baseline resolution, but it should not be described as mesh-independent until the medium and fine solutions are compared.

TODO: Missing figure
`mesh_independence_throat_mach.png`
 Generate from computed CFD output before final portfolio export.

Figure 7: Mesh-independence plot for throat Mach number. If the cases have not been solved, the figure is a generated placeholder indicating missing computed data.

12 Visual CFD Results

The repository includes a ParaView export guide at `docs/paraview_export_guide.md`. The intended visual result set for the portfolio PDF is:

- full-nozzle Mach contour;
- pressure contour;
- temperature contour;
- mesh zoom near the throat;
- shock-region zoom for the internal-shock case if present;
- centerline plots generated by Python validation scripts.

No synthetic contour screenshots are used in this report. The choked-case contour and mesh figures shown here were exported from the regenerated OpenFOAM solution.

TODO: Missing figure
`mesh_independence_mass_flow.png`
Generate from computed CFD output before final portfolio export.

Figure 8: Mesh-independence plot for outlet mass flow. If the cases have not been solved, the figure is a generated placeholder indicating missing computed data.

TODO: Missing figure
`choked_mach_contour.png`
Generate from computed CFD output before final portfolio export.

Figure 9: Full-nozzle Mach contour for the regenerated choked case, exported from ParaView.

13 Repository Reproducibility

The repository is organized so that source files and generated outputs are separated. The case dictionaries, scripts, documentation, and report source are intended to be version-controlled. Generated OpenFOAM time directories, logs, `postProcessing`, processor directories, mesh files under `constant/polyMesh`, Python bytecode, and LaTeX auxiliary files are ignored.

Important reproducibility choices include:

- `blockMeshDict` is stored in `system/blockMeshDict`, so meshes are regenerated from a clean checkout;
- validation scripts parse OpenFOAM ASCII fields directly and do not require ParaView;
- mass-flow validation does not assume `phi` exists;
- mesh-independence studies leave observed results blank until computed;
- `Allrun`, `Allclean`, and `Allvalidate` accept selected case directories.

```
./Allrun cases/choked
./Allvalidate cases/choked
python3 scripts/mesh_independence.py
```

14 Limitations

The current case is intentionally compact and validation-focused. Important limitations are:

- The mesh is quasi-two-dimensional, not a full axisymmetric or three-dimensional nozzle.
- Slip walls suppress viscous boundary-layer effects and wall shear stress.
- The transport model uses constant viscosity rather than Sutherland’s law.

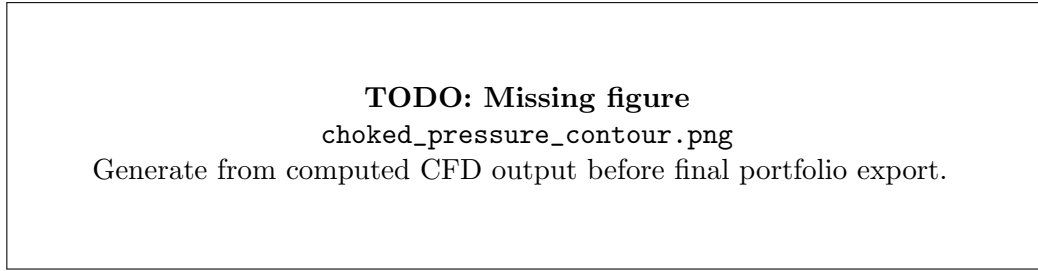


Figure 10: Full-nozzle pressure contour for the regenerated choked case, exported from ParaView.

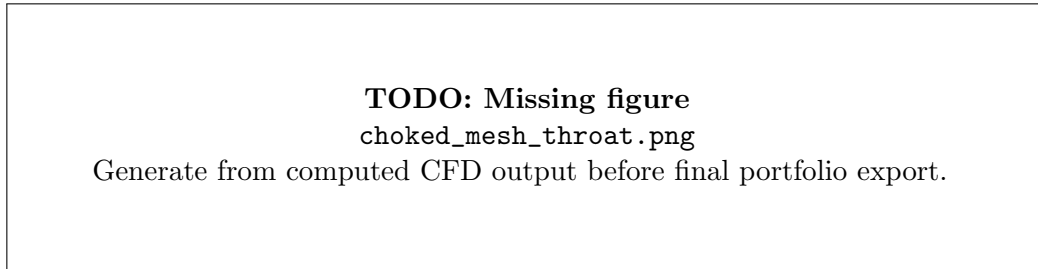


Figure 11: Mesh zoom near the throat for the regenerated choked case, exported from ParaView.

- The solution is transient and run to a finite physical time; additional monitoring of mass flow and residual trends is needed before claiming a fully steady state.
- The repository contains subsonic, choked, and internal-shock case variants; all three pressure-ratio cases now have regenerated validation data, while mesh-independence cases still require separate runs before claiming quantitative grid convergence.
- Mesh-independence plots currently act as a study framework unless the corresponding CFD cases have been solved.
- ParaView contour figures are available for the regenerated choked case; additional case variants still require separate exports after their CFD runs.
- One-dimensional isentropic theory does not model viscous losses, multidimensional effects, shock thickness, numerical diffusion, or total-pressure loss across shocks.

15 Portfolio Value

This project demonstrates the complete CFD workflow for a canonical aerospace application: geometry definition, structured meshing, compressible-flow solver setup, stability control through Courant-number management, post-processing, and comparison against analytical theory. The case is small enough to run on a laptop while still showing key high-speed flow concepts such as acceleration through a throat, pressure-temperature coupling, and the role of pressure ratio in choking.

16 Appendix: Generated PNG Figure Gallery

This appendix intentionally repeats the lightweight PNG outputs that are currently present in `docs/images`. It ensures the portfolio PDF ends with the available visual evidence from the regenerated choked run, while retaining explicit placeholders only for studies that have not yet been solved.

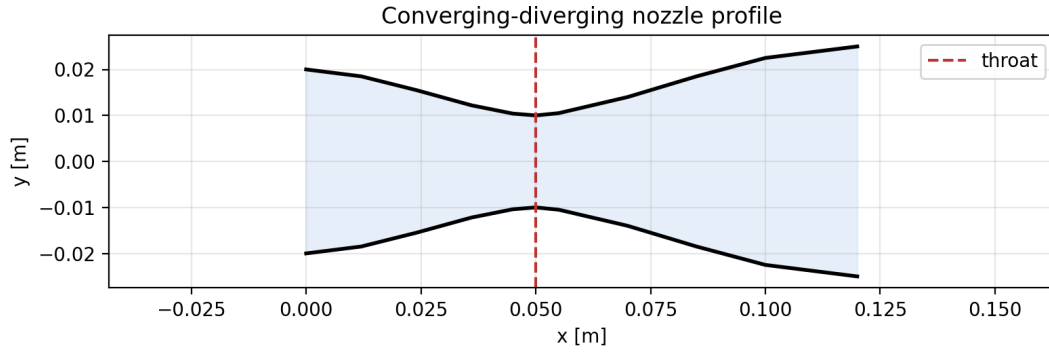


Figure 12: Generated nozzle geometry plot.

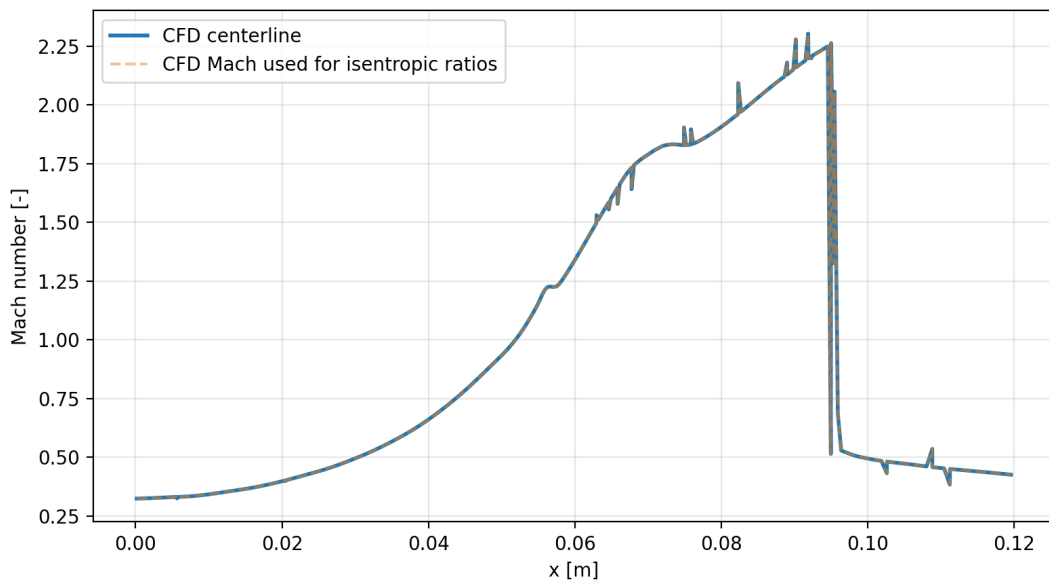


Figure 13: Generated centerline Mach plot.

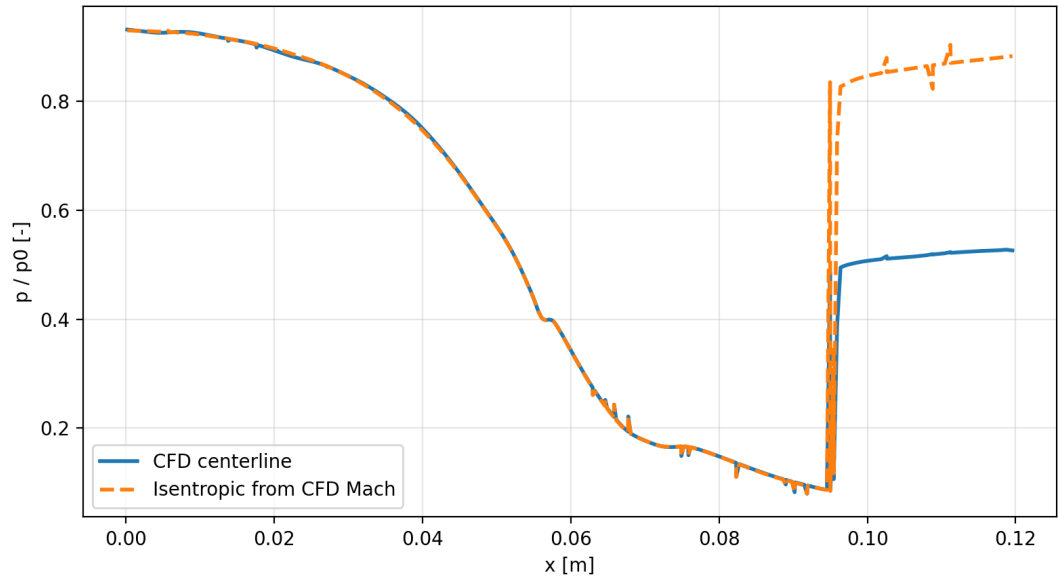


Figure 14: Generated centerline pressure-ratio plot.

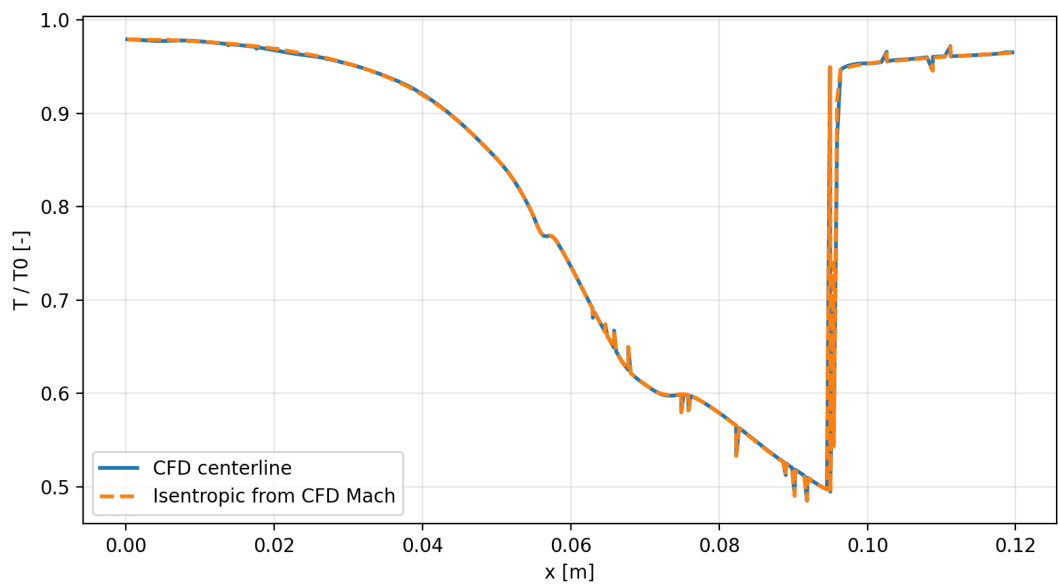


Figure 15: Generated centerline temperature-ratio plot.

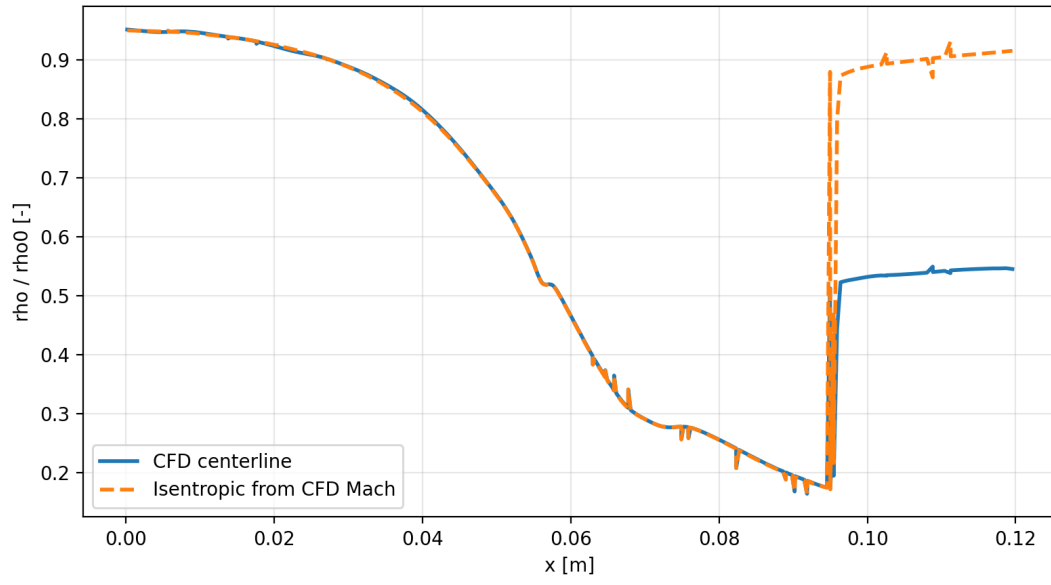


Figure 16: Generated centerline density-ratio plot.

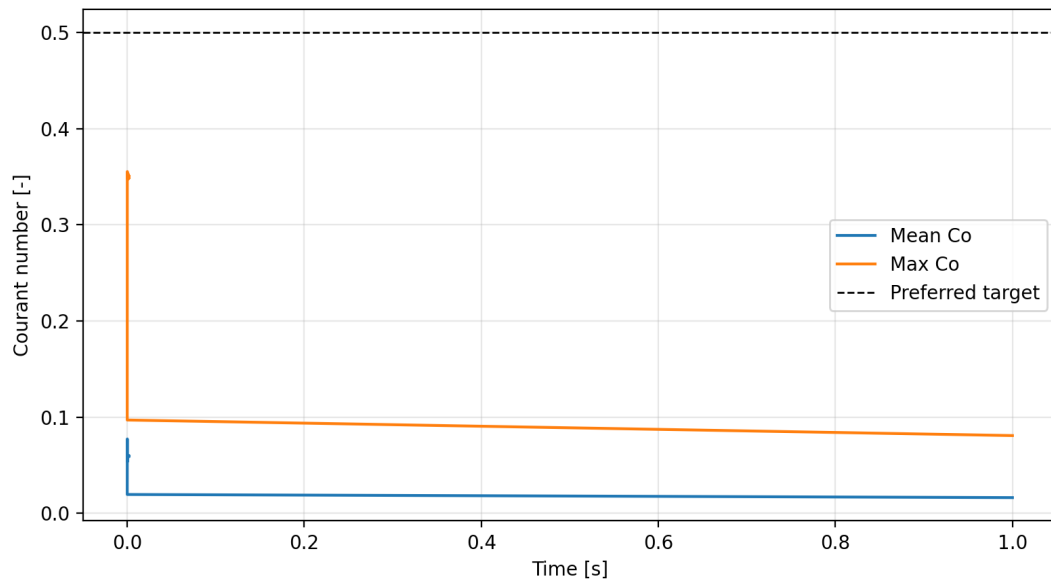


Figure 17: Generated Courant-number history plot.

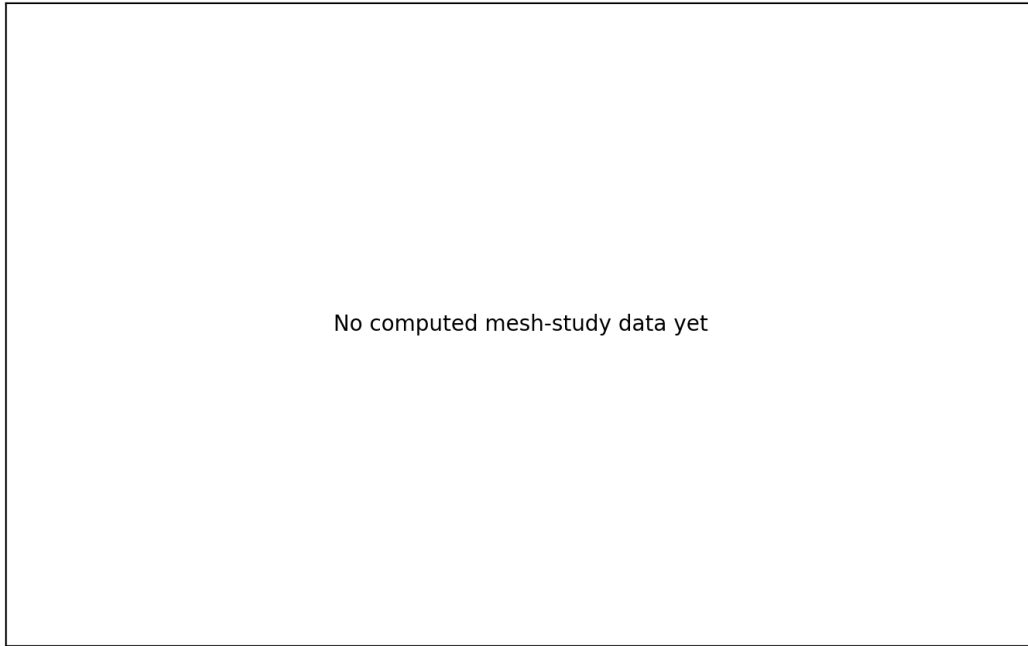


Figure 18: Mesh-independence throat-Mach plot. If the mesh cases have not been solved, this image is a generated missing-data placeholder.

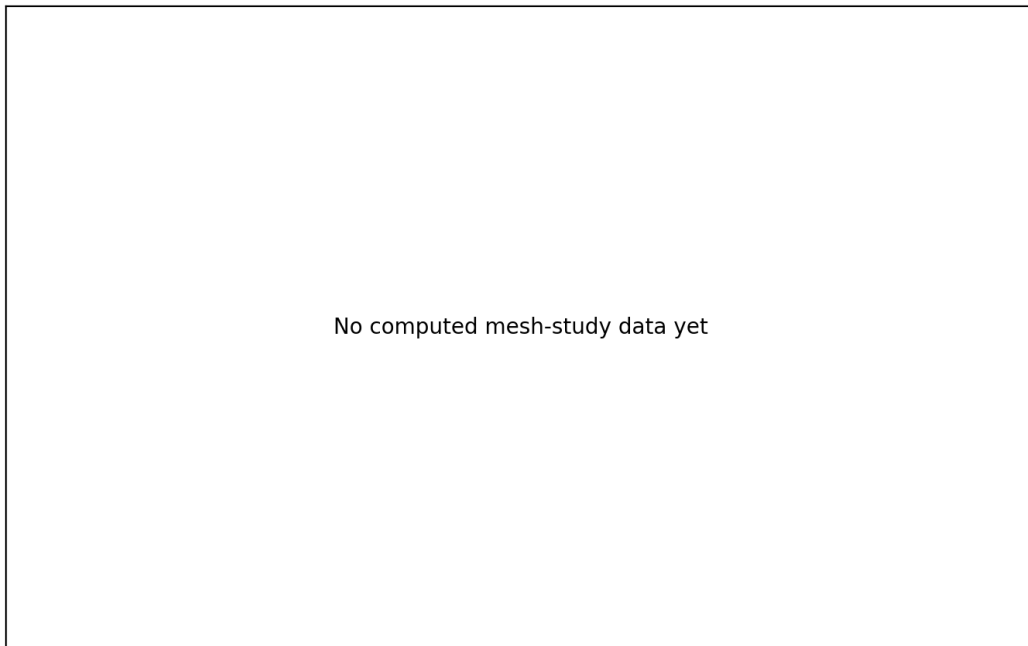


Figure 19: Mesh-independence mass-flow plot. If the mesh cases have not been solved, this image is a generated missing-data placeholder.

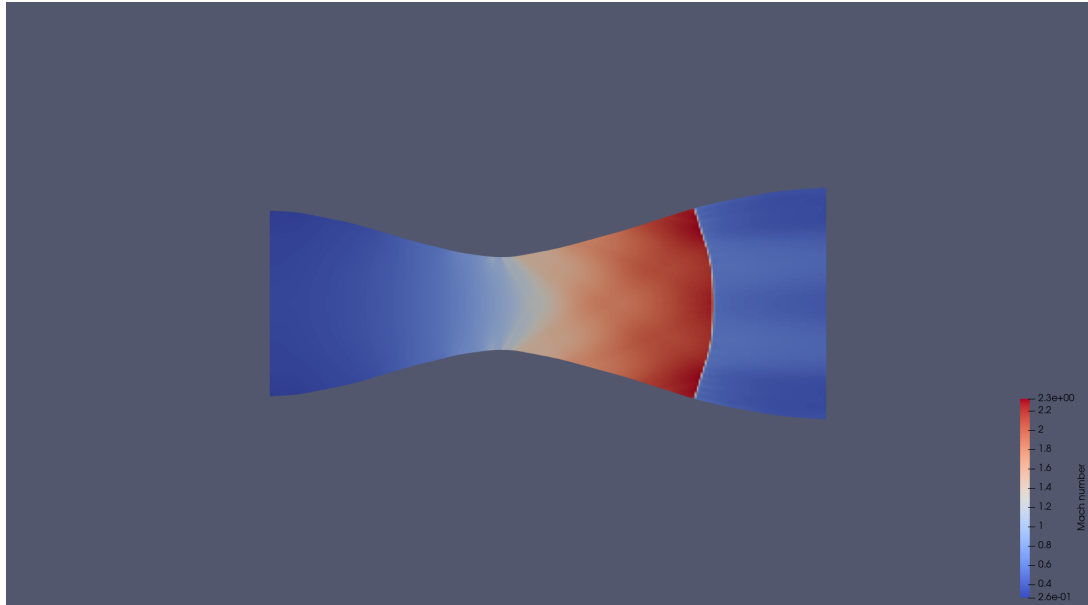


Figure 20: ParaView Mach contour for the regenerated choked case.

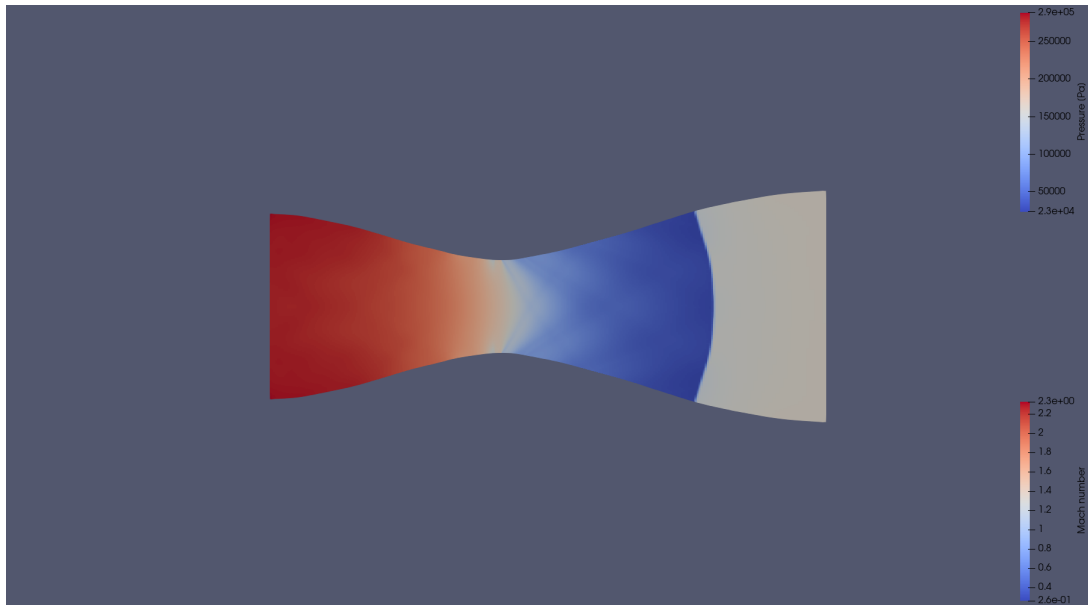


Figure 21: ParaView pressure contour for the regenerated choked case.

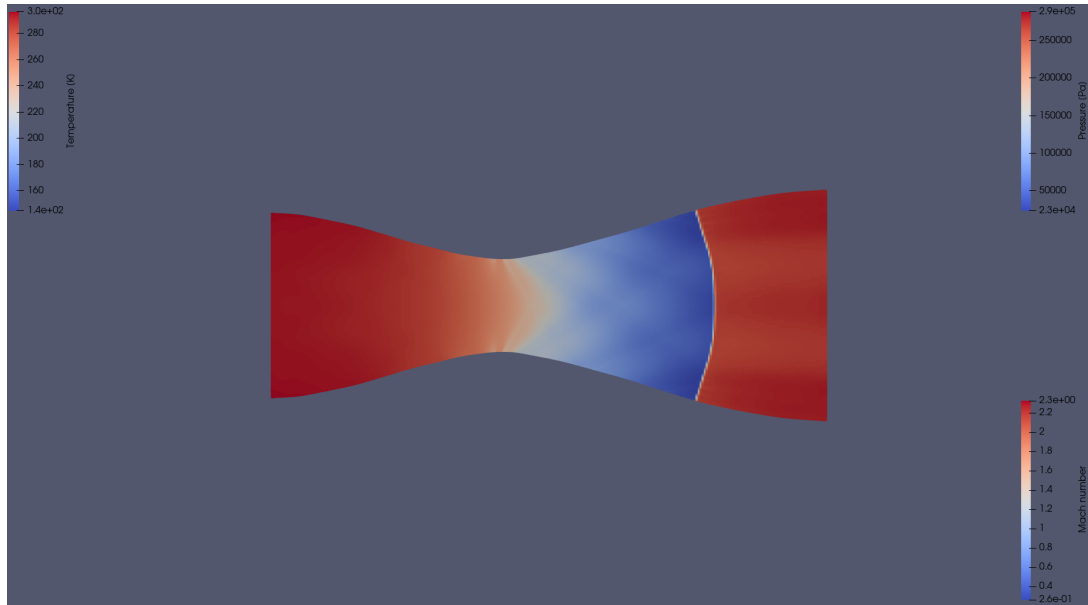


Figure 22: ParaView temperature contour for the regenerated choked case.

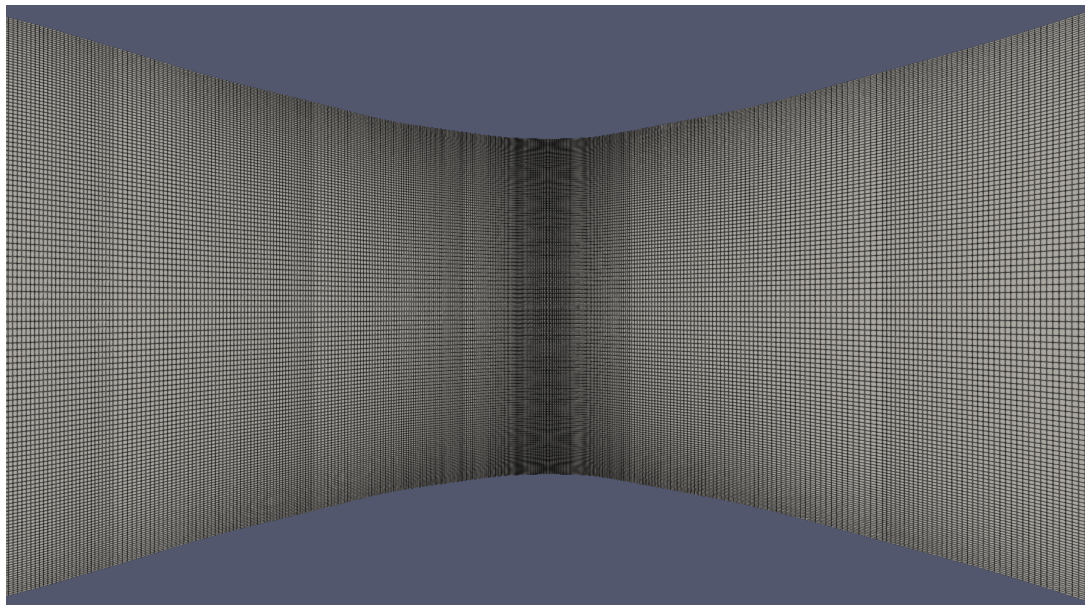


Figure 23: ParaView mesh zoom near the throat for the regenerated choked case.